Project

Circuits and Systems for Radio-Frequency

Design of a Low-IF RF-CMOS front-end for a Low Data Rate SDR

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1- Introduction

Starting from the reference architecture associated with the concept of Software Defined Radio (SDR), the project aims to study, design and simulate a complete narrowband Low-IF RF front-end (RFFE), depicted in Fig. 1, using a 65 nm CMOS technology.

The objective is not only to design the individual building blocks, but also to understand how their impedances, gain, noise, linearity and frequency-selective behaviour interact at system level. Particular emphasis is placed on engineering reasoning, design trade-offs and the interpretation of simulation results.

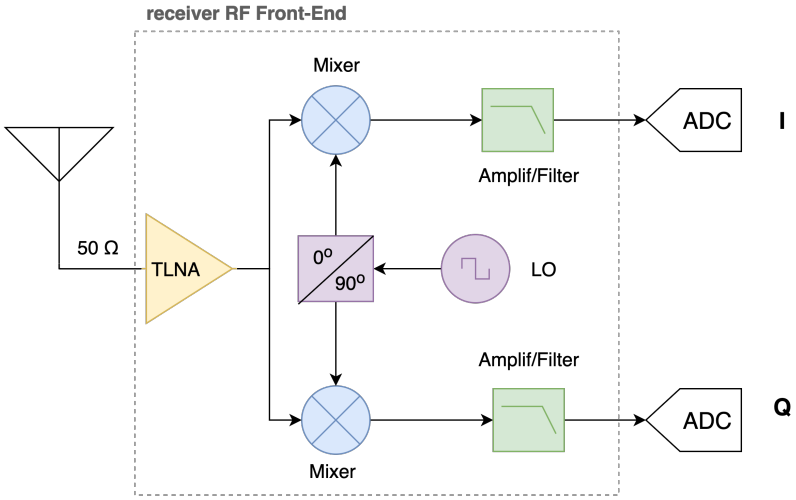


Figure 1- Simplified block diagram of a Receiver RF Front-End for a SDR.

The design should start from the receiver high-level requirements and proceed to a consistent allocation of specifications to the individual blocks. The selected circuit parameters must be justified from first-order analysis and subsequently validated and refined using Cadence/SpectreRF simulations.

The reference RFFE architecture to be implemented includes:

- Low-noise amplifier (LNA): single-ended, narrowband, inductively-degenerated common-source topology;
- I/Q mixer: current-mode passive CMOS switching mixer;
- Baseband transimpedance stage: low-input-impedance differential TIA with first-order RC filtering;
- Low-IF channel filter: second-order low-pass filtering for channel selection and anti-aliasing, followed by additional gain if required;
- Quadrature LO drive for the I/Q mixer. Ideal LO sources may be used in the core receiver design; the required LO amplitude, waveform and buffer/driver capability shall be specified.

2- Specifications / design requirements and constraints

- TX EIRP: 0 dBm
- Coverage range (distance): up to 100 m
- Modulation: FSK/ASK, 40kbps and BER=10^{-4}. The BER requirement is used for receiver-level sensitivity/noise budget initial analysis; implementation and simulation of the digital demodulator are outside the scope of this project.
- ISM band: 868MHz or 915MHz.
- Narrowband approach.
- Low-IF architecture. The Low-IF frequency will be assigned to each group; values in the approximate 100kHz to 300 kHz range are targeted.
- Receiver linearity reference for system-level budgeting: P1dB > -10 dBm. This is a design target rather than a strict pass/fail requirement; deviations shall be technically justified.
- Receiver linearity reference for system-level budgeting: IIP3 >= -14 dBm. This is a design target rather than a strict pass/fail requirement; deviations shall be technically justified.
- Reference blocker scenario for system-level analysis: a single-tone blocker at f_{IF} + 300 kHz with −30 dBm at the antenna, with the wanted signal at P_{min} + 6 dB. A detailed full-receiver blocker simulation is not mandatory, but recommended; the scenario should be used to discuss filtering, linearity and ADC headroom.
- ADC specs: input full scale range 1Vpp, 8 bits, sampling frequency of 10 MHz
- Technology: CMOS 65 nm
- Nominal supply voltage: VDD = 1.2 V. A lower supply voltage may be adopted if technically justified.
- Inductors, resistors, and capacitors from the 65 nm design kit
    - Consider an approximate model of the on-chip inductor with parasitic capacitances; CP of 10 fF; also, consider an on-chip inductor quality factor of 7. You can fine-tune these values by checking the 65 nm design kit;
    - Due to area constraints, limit the total inductance;
    - On-chip capacitor with a bottom-plate parasitic capacitance of 4%.

To reduce direct reuse of previous designs while preserving comparable difficulty, selected parameters (for example RF band, Low-IF frequency and blocker offset) may be assigned differently to different groups. All groups will implement the same reference architecture and will be assessed using the same technical criteria. The values assigned to each group will be provided.

If one or more specifications cannot be fully achieved during the design/simulation, provide a technical explanation and propose an updated value based on your analysis/results.

3- Design and Validation

65 nm CMOS technology Design and simulation process

The design process should include the determination of the relevant circuit equations, first-order models and/or design tables from which device and component values are obtained (e.g., W, L, capacitance, inductance and resistance). The operation and performance of the designed circuits shall then be confirmed through SpectreRF (Cadence) simulation using the available BSIM4 transistor

models. Transistors processing RF signals should use the RF-oriented devices available in the 65 nm component library when appropriate. The SpectreRF environment includes Periodic Steady State (PSS) and related analyses suitable for periodically time-varying circuits such as mixers. The validation plan should remain compact and use each analysis only where it provides useful design information. It should include, where applicable:

- Selected distortion/nonlinear analyses, including P1dB and/or IIP3 for the blocks where they are most relevant;
- Noise simulation (including PNOISE where appropriate);
- At least one meaningful PVT analysis for one or more building blocks.

Create a compact and reusable set of testbenches that verifies the relevant performance of the main building blocks and the end-to-end operation of the receiver. PSS/PAC/PNOISE analyses should be used where they are robust and technically appropriate, particularly for individual periodically time-varying blocks such as the mixer. A complete-receiver PSS simulation is not required. Transient simulation is the recommended method for end-to-end functional validation. The objective is to obtain a small number of technically meaningful results, not an exhaustive set of plots.

The final report should be concise and should document the analysis, design decisions and most relevant simulation results. It should clearly include:

- A compact receiver-level specification/budget and the resulting requirements for the main building blocks (gain, noise, linearity, bandwidth and power consumption; estimated area where meaningful);
- First-order analysis supporting the main design choices, including the most relevant noise and linearity considerations;
- High-level calculations or simulations (Python, Octave, MATLAB or equivalent), when they add useful insight;
- Electrical simulations using realistic device and passive-component models;
- A concise comparison between required, analytically predicted and simulated performance, including justified deviations from the original targets;
- At least one PVT result for a technically relevant block (for example the LNA or TIA), rather than a large collection of undiscussed simulation plots.

The following requirements clarify the expected work for each part of the receiver.

For the LNA

- Topology: single-ended narrowband common-source LNA with inductive source degeneration;
- Justify the selected transistor operating region and device sizing;
- Input: single-ended, designed for the specified source impedance;
- Output: current-mode interface suitable for driving the following passive mixer;
    - Use the narrowband output network and the real on-chip inductor model to establish the required RF load and interface to the mixer;
    - Discuss explicitly how the LNA output impedance/network interacts with the passive mixer and baseband load;
- Noise figure

- Gain/transconductance: identify the most meaningful quantity for the LNA-to-mixer current-mode interface;
    - Input matching: design for the specified source impedance and report S11 over the relevant RF band;
- Power consumption
- 1 dB compression point as supporting large-signal characterization; an extensive sweep is not required if IIP3 and the receiver linearity budget are adequately established;
- Linearity: IIP3 as the primary LNA nonlinearity metric;
- Estimated area

For Mixer (including the TIA)

- Topology and operation:
    - Current-driven passive CMOS switching mixer (required);
    - Quadrature LO drive with the switch sizing and required LO amplitude/buffer capability explicitly justified;
- Conversion Gain;
- Linearity: IIP3; a separate mixer/TIA P1dB extraction is not required unless it provides useful additional insight;
- Noise analysis, using the most appropriate SpectreRF method for the selected testbench;
- TIA: use a low-input-impedance transimpedance stage. A differential TIA with RC feedback is the reference solution and shall include first-order low-pass filtering; two matched single-ended TIAs may alternatively be used if the differential operation and common-mode requirements are properly addressed;
- Characterize the mixer/TIA interface, including conversion gain and the effect of the baseband input impedance. A short high-side/low-side comparison is encouraged when it can be obtained without excessive simulation effort;
- Power consumption;
- Estimated area.

Recommended reading: A. Mirzaei et al., “Analysis and Optimization of Current-Driven Passive Mixers in Narrowband Direct-Conversion Receivers,” IEEE JSSC, 2009. Focus on the physical implications for the mixer/TIA interface; the full derivation is not required.

For the Low-IF baseband chain

- Provide the gain required after the TIA while respecting the ADC input range;
- Implement a second-order low-pass filter for channel selection and anti-aliasing before the ADC;
- Two matched single-ended Sallen-Key sections, one per differential branch, are acceptable for simplicity;
- A differential active-RC implementation, such as a Rauch / multiple-feedback topology, is also acceptable and may be preferred when properly justified;
- Verify the relevant passband gain, cut-off frequency and attenuation at selected out-of-band frequencies relevant to anti-aliasing;
- Discuss differential signal handling and common-mode levels at the TIA/filter/ADC interfaces;

- Evaluate power consumption and the noise/linearity quantities that are relevant to the receiver budget; an exhaustive characterization of the baseband filter is not required.
- Additional gain stages may be used only if required by the receiver-level gain budget.

For the full front-end receiver

The complete RF front-end shall be validated as an integrated receiver chain using an ideal quadrature LO if desired. The objective of this simulation is functional validation rather than exhaustive receiver characterization. A complete-chain PSS simulation is not required. A small number of transient simulations should be used to demonstrate:

- RF-to-Low-IF conversion in the I and Q paths for a representative wanted input signal;
- Approximate overall gain and consistency with the receiver-level gain budget;
- Appropriate signal levels through the chain and no clipping, particularly at the ADC interface;
- The expected Low-IF output spectrum using a simple FFT or equivalent spectral view;
- Consistency of the integrated response with the most important block-level results. Receiver NF, P1dB and IIP3 may be estimated from the block-level results and receiver budget; direct full-chain extraction is not required.
- Total power consumption obtained from the implemented blocks;
- A brief discussion of the high-side/low-side and image-frequency implications of the Low-IF/IQ architecture. A dedicated full-receiver simulation of this effect is not mandatory if it has already been illustrated at mixer/TIA level.

Detailed full-receiver extraction of NF, P1dB, IIP3 and blocker tolerance is not mandatory. These quantities should instead be supported, where relevant, by block-level simulations and first-order receiver budgeting. The purpose of the integrated simulation is to demonstrate that the designed receiver works as a coherent signal chain. Quality of engineering decisions and interpretation of results are valued more than the number of simulation plots.
More extensive full-receiver nonlinear or blocker simulations are optional and may be included only when they add clear technical insight and can be obtained with reasonable simulation effort.

Additional task for groups of four students

- Groups of four students shall additionally design an I/Q local-oscillator subsystem, using a ring oscillator/divider or another justified approach. For groups of three students, the LO itself is not part of the required transistor-level design and ideal quadrature LO sources may be used. The additional LO task for groups of four compensates for the larger group size and does not constitute an automatic grading bonus;
- For the additional LO task, design the output drivers/buffers required to drive the passive mixer switches and verify the generated quadrature phases. All groups shall nevertheless specify the LO amplitude and driver requirements imposed by their mixer;
- Estimate the additional power consumption and area of the LO subsystem.

4- LNA Layout (optional)

The LNA layout, including the real inductors available in the process design kit (PDK), is optional. When attempted, it should be used to discuss realistic RF layout constraints and expected differences between schematic-level and post-layout behaviour.

5- Evaluation

The RF front-end shall be analysed, designed and validated with emphasis on the main RF/Low-IF blocks and on end-to-end functionality. The final technical report must not exceed 10 pages, including figures and tables, but excluding the cover page and references. The report should be concise: extended schematics, raw simulation data, calculation files and scripts should be delivered separately when useful, rather than used to increase the report length. CAD laboratory access is planned until 18 December 2026; the report will be submitted during the first week of January 2027 (exact deadline on Moodle), followed by individual discussions.

- Deliverables: technical report, relevant calculation/simulation files (Python, Octave, MATLAB, Excel or equivalent) and the complete Cadence design database;
- Individual discussion: each student must be able to explain and justify the submitted design choices, calculations, schematics, testbenches and simulation results.

Project classification:

NPRJ = 35% NTECH + 15% NREL + 50% NDISC

NTECH evaluates the technical quality and rigour of the design and simulations; NREL evaluates the clarity, conciseness and quality of the technical report; NDISC is the individual discussion grade and may therefore differ among members of the same group. The project grade NPRJ contributes 60% to the final CSRF course grade.

Assessment criteria:

- NTECH considers correctness and completeness of the implemented receiver, quality of first-order design and receiver budgeting, appropriateness of the simulation/testbench methodology, quality of trade-off analysis, and consistency between the RF, mixer/TIA and Low-IF stages.
- NDISC considers individual understanding of circuit operation, ability to justify design choices, understanding of simulation methods and limitations, ability to interpret and predict circuit behaviour, and ability to reason about modifications or operating conditions not explicitly presented in the report.
- As a general grading reference: 10-13 indicates a basically functional but incomplete or weakly justified design; 14-16 a correct and coherent project with the main requirements achieved or justified; 17-18 a very solid project with clear trade-offs, good theory/simulation correlation and strong system understanding; 19-20 exceptional technical maturity, optimization and critical insight.

A complete and functional design corresponds to a good project. A high grade does not require unnecessary circuit complexity or a larger number of simulations. Grades above 17 require clear evidence of strong engineering insight, critical analysis and individual mastery of the receiver; exceptional grades require exceptional engineering understanding, not additional circuit complexity.